

TeamName_Finance_Module1.zip

- | — /prototype/
 - | — (Your complete FastAPI backend code)
 - | — (Your complete React frontend code)
 - | — README.md (Setup instructions, one-line flow description, notes)
- | — Build-Log.pdf (or .docx)
 - | — Entry 1: Theme, Problem Statement, Intended Build, Leverage Map
 - | — Entry 2: 4D Check table and delegation map
 - | — Entry 3: Six-part brief, output, verified claim & source, tagged claims
 - | — Entry 4: Unwind, reframes/objections, 3 ranked assumptions, Problem Statement v2
 - | — Entry 5: Hypothesis, ICP, substitute, mechanism, sized opportunity, insight ledger, chosen flow
- | — Entry 6: PROJECT_SPEC.md, tech stack justification, stranger test notes, reusable workflow
- | — Build-Summary.md (or .txt)
 - | — 1-page summary for an outside reader (Theme, Problem v2, What was built, Tech stack, Evidence vs Assumptions, Next flow)

Build Summary: FactorPulse

Theme

Fintech (hackFront Innovation Theme)

Problem Statement

* Final Version: Micro, Small, and Medium Enterprises (MSMEs) face chronic working capital paralysis because mid-market enterprise buyers enforce 60-to-90-day payment terms, locking receivables required for weekly payroll and raw material inventory. Traditional underwriting takes 7 to 14 days of manual credit assessment, shutting out vendors who lack physical collateral or audited balance sheets.

* Evolution: Our initial statement focused broadly on freelance receipt parsing and manual expense tracking, which was an administrative productivity tool. We pivoted to B2B counterparty transaction risk underwriting to solve structural working-capital lockups in vendor supply chains.

What We Built

* The Flow: An automated B2B invoice risk underwriter and instant liquidity engine. An MSME enters invoice parameters (amount, buyer credit tier, tenor, and past payment delay). The system calculates a deterministic counterparty credit risk score, determines an allowable advance rate (70%–90%), calculates a dynamic discount fee, and issues an instant capital advance offer with an automated underwriting credit memorandum.

* What It Deliberately Does Not Do: It does not execute live automated bank disbursements, does not connect to live Indian GST e-invoicing portals (mock validation is used), and does not support multi-currency foreign exchange invoicing.

Tech Stack

* Backend: Python & FastAPI. Chosen for high-speed deterministic financial math, strict Pydantic payload validation, and clean REST endpoints.

* Intelligence Layer: Google Gemini API. Used strictly for contextual synthesis (generating structured underwriting credit memorandums and counterparty risk narratives from quantitative scores).

* Frontend: React (Vite) + Tailwind CSS. Chosen for rapid client-side state responsiveness, interactive LTV/tenor liquidity sliders, and zero-latency UI updates.

Evidence Position

* Proven: Deterministic actuarial formulas can price risk and calculate dynamic capital discounts in milliseconds without third-party API dependencies. The spread between buyer creditworthiness and tenor mathematically offsets liquidity risk.

* Unverified Assumption: We assume mid-market enterprise buyers will legally accept third-party notice of assignment (NOA) and route invoice payouts into an escrow account without disputing delivery quality.

What We Would Build Next

* Next Flow: Direct integration with GSTN (Goods and Services Tax Network) to verify Invoice Reference Numbers (IRN) and e-Way bill data programmatically.

* Prerequisite: Requires regulatory sandbox access and commercial GST Suvidha Provider (GSP) API credentials.

Complete hackFront Cohort Build Log

Build Log, Entry 1: Problem Definition & Leverage Map

Theme

Fintech (hackFront Innovation Theme)

Problem Statement

Micro, Small, and Medium Enterprises (MSMEs) face severe operational cash-flow gaps because enterprise buyers enforce 60-to-90-day payment cycles, trapping working capital required for weekly payroll and raw inventory. When these suppliers seek invoice discounting to access liquidity, traditional underwriting requires 7 to 14 days of manual credit assessment, paperwork, and balance-sheet vetting. This delay costs growing suppliers lost contracts and pushes them toward high-interest informal debt. Solving this now is critical because digital invoicing and GST data are standardized across India, yet automated, real-time invoice risk underwriting remains inaccessible to Tier-2 and Tier-3 vendors.

Intended Build

We intend to build **FactorPulse**, an automated B2B invoice risk underwriter and instant liquidity engine built with a FastAPI backend and React frontend. An MSME vendor submits pending invoice parameters (buyer tier, tenor, invoice value, and historical payment delay metrics) to receive an instant risk score, dynamic discount rate, and approved early cash advance. The system produces an immediate capital advance offer and an audit-ready financial term sheet in seconds rather than weeks.

Leverage Map

#	Major Task	Mode	Rationale
1	FastAPI Core & REST Endpoint Scaffolding	Automation	Standard boilerplate endpoints, CORS rules, and Pydantic validation schemas generated directly via templates.
2	Actuarial Risk & Discount Pricing Logic	Augmentation	Formulated deterministic risk equations linking tenor and buyer default probability, assisted by LLMs for parameter edge cases.
3	AI Underwriter Credit Memorandum	Augmentation	Structured LLM prompt generating a concise credit memorandum detailing counterparty strength, industry risk, and default covenants.
4	React Visual Underwriting Console	Automation	UI scaffolding using Tailwind to display real-time metric cards, cash advance totals, and fee breakdowns.
5	Interactive Liquidity & Tenor Controls	Automation	Dynamic client state updating net cash unlocked and factoring fees as the user adjusts financing parameters.
6	Automated Factoring Term Sheet Generator	Agency	An autonomous agent structures and formats an institutional factoring agreement with customized default covenants.

What Stays Ours

- The Underwriting Matrix & Risk Thresholds:** We will not delegate credit decision boundaries to AI; the exact cutoff conditions (such as rejecting tenors beyond 90 days from unrated buyers) remain strictly governed by our hardcoded risk formula.

- **The Business Model & Margin Economics:** The factoring fee spread (2%–4%) and reserve holdback percentages are our deliberate financial decisions to ensure the platform operates as a commercially viable lending engine.

Build Log, Entry 2: The 4D Check Table & Delegation Map

The 4D Check Table

Task Dimension	Description	Human or AI Dominant	Safety Boundary / Guardrail
Data Integrity	Validating financial numerical inputs (invoice value, tenor, past delay).	Human (Pydantic deterministic validation)	Invoices with negative amounts or tenors exceeding 120 days trigger hard rejection before any LLM execution.
Domain Logic	Pricing formula linking repayment delay to dynamic APR fees.	Human	Risk calculations use explicit formulas: $\text{Fee} = \text{Principal} * (\text{Base} + \text{Spread}) * (\text{Tenor} / 360)$.
Delivery Tone	Drafting the credit memorandum and executive factoring term sheet.	AI (Augmentation)	Prompt output constrained to strict JSON fields (summary, covenants, rejection_reason).
Decision Authority	Final approval/rejection verdict of the credit advance.	Human	LLMs provide rationale only; the system enforces strict score cutoffs (Score < 50 = Reject).

Delegation Map

- **Delegated to AI:** Natural language synthesis of the credit memo, formatting the term sheet agreement, generating mock test scenarios, and writing UI CSS utility classes.

- **Retained by Humans:** Actuarial formula constants, underwriting cutoff thresholds, API route boundaries, and core state management logic.

Build Log, Entry 3: Six-Part Brief & Verified Claims

The Six-Part Brief

1. **Context:** FactorPulse is an instant invoice risk underwriting engine for MSMEs facing 60–90 day payment terms.
2. **Task:** Ingest an invoice payload, calculate counterparty credit score (0–100), dynamic LTV, discount fee, and generate a 3-sentence credit memo.
3. **Constraints:** No third-party financial API dependencies. Math must be deterministic. Zero hallucinations on numerical outputs.
4. **Input Data Schema:** JSON object containing `invoice_amount`, `buyer_tier`, `tenor_days`, `past_delay_days`, and `industry`.
5. **Output Data Schema:** Strict JSON containing `risk_score`, `risk_tier`, `advance_rate`, `net_advance_amount`, `discount_fee`, `reserve_amount`, and `credit_memo`.
6. **Evaluation Criteria:** Latency under 500ms, zero NaN outputs, exact penny reconciliation: `net_advance + discount_fee + reserve == invoice_amount`.

Verified Claim & Source

- **Claim:** Indian MSMEs face over ₹10.7 lakh crore (\$130 billion) in delayed payments from corporate buyers, with settlement periods averaging 60–90 days.
- **Source:** Global Alliance for Mass Entrepreneurship (GAME) & Dun & Bradstreet Research Report on Delayed Payments to Indian MSMEs.

Tagged Claims List

- *[Verified]* Section 43B(h) of the Indian Income Tax Act disallows tax deductions for enterprise expenses unless payments to registered MSMEs are settled within 45 days.
- *[Verified]* RBI TReDS platforms handle over ₹1,00,000 crore annually in discounted invoice volume.
- *[Assumption]* Mid-market buyers will consent to third-party factoring without legal counter-claims on delivery quality.
- *[Assumption]* An MSME will accept a 2.5% discount fee to receive liquidity in 5 minutes rather than waiting 90 days.

Build Log, Entry 4: Unwind, Reframes, Critic Objections & Problem Statement v2

The Unwind & Reframes

- Initial Frame: "MSMEs need a better expense tracker to know where money goes."
- The Unwind: Expense tracking addresses the wrong side of the balance sheet. MSMEs fail not from excessive tracking friction, but from working capital lockups in unpaid accounts receivable.
- Reframe: "MSMEs do not need budgeting software; they need instant liquidity against verified customer receivables."

Three Ranked Assumptions & 48-Hour Tests

Rank	Critical Assumption	48-Hour Validation Test	Metric for Success
1	Lenders can safely price invoice risk using buyer metadata without full supplier balance-sheet auditing.	Review 20 historical factoring default datasets from open NBFC portfolios.	Default rate correlation with buyer credit tier > 0.82.
2	Suppliers are willing to sacrifice 2%–3% gross margin for day-1 cash advances.	Interview 5 local hardware/packaging vendors on cash discount preferences.	At least 4 out of 5 vendors confirm taking early cash discounts.
3	Large corporate buyers will pay into third-party escrow accounts upon invoice maturity.	Review legal enforceability of Notice of Assignment (NOA) under Indian Factoring Regulation Act 2011.	Clear legal precedent established under Section 7 of the Act.

Critic Objections & Rebuttals

- **Critic:** "TReDS and banks already do invoice discounting. Why build FactorPulse?"
- **Rebuttal:** TReDS onboarding requires weeks of institutional KYC, limiting its use to Fortune 500 corporate buyers. FactorPulse provides algorithmic micro-underwriting for mid-market enterprises (₹10 Cr–₹100 Cr revenue) that banks ignore.

Problem Statement v2

Micro, Small, and Medium Enterprises (MSMEs) face severe operational cash-flow gaps because mid-market buyers enforce 60-to-90-day payment cycles, trapping working capital required for weekly payroll and raw inventory. Traditional underwriting requires 7 to 14 days of manual credit assessment, shutting out vendors who lack physical collateral or audited balance sheets. This delay costs suppliers supply-chain stalls and forces them into predatory informal credit. Solving this now is critical because Indian supply chains are digitizing rapidly, creating an opportunity for real-time, algorithmic counterparty risk pricing.

Build Log, Entry 5: Persona, Hypothesis & Chosen Flow

Ideal Customer Profile (ICP)

- **Entity:** Tier-2/Tier-3 component manufacturers, packaging vendors, and IT staffing agencies in India.
- **Annual Turnover:** ₹1 Crore to ₹15 Crores.
- **Pain Point:** Healthy order books and high profitability, but constant cash crunches due to 60-to-90-day buyer payment terms.

Hypothesis

If an MSME supplier can receive an algorithmic credit score and binding discount advance offer within 60 seconds of submitting basic invoice parameters, they will adopt invoice factoring over predatory informal credit lines.

Existing Substitutes

- High-interest uncollateralized business loans (18%–36% APR).
- Manual bank overdraft facilities requiring commercial property collateral.
- Informally begging corporate clients for early release with ad-hoc manual discounts.

Sized Opportunity

Over 63 million MSMEs contribute ~30% of India's GDP. Delayed payments lock up ~₹10.7 lakh crore annually. Capturing even 0.05% of this liquidity pool through micro-discounting represents a ₹535 crore financing opportunity.

The Chosen Flow

The Instant Factoring Underwrite Flow:

1. Supplier inputs invoice value (₹), buyer corporate tier, payment tenor (days), and historical payment delay.
2. System validates data through Pydantic schemas and passes it to the Actuarial Risk Engine.
3. System calculates counterparty risk score, LTV advance percentage, factoring fee, and instant payout amount.
4. Gemini generates a 3-sentence credit underwriting memorandum summarizing counterparty stability and default covenants.
5. React UI displays an immediate "Cash Advance Available" card with dynamic LTV sliders and a term sheet download option.

Build Log, Entry 6: PROJECT_SPEC.md, Tech Stack & Workflow

Technical Specification ([PROJECT_SPEC.md](#))

- **Backend:** FastAPI (Python 3.11+), Uvicorn server, Pydantic v2.
- **AI Model:** Google Gemini (via [google-genai](#) SDK) for qualitative credit memos.
- **Frontend:** React 18, Vite, Tailwind CSS, Lucide-React.
- **Architecture Pattern:** Decoupled client-server architecture via standard REST JSON APIs.

Stranger Test Notes

- **Test Subject:** Tested with an external engineering student with zero finance background.
- **Observation:** The tester immediately understood the value proposition once the "Instant Payout" metric card changed in real time when sliding the tenor from 30 to 90 days.
- **Feedback Incorporated:** Added an explicit fee breakdown label showing that the reserve is returned once the client settles the bill.

[Client Form Input]

—(POST /api/v1/underwrite)—> [Pydantic Input Sanitization]



[Deterministic Risk Formula]
(Score: 0-100, Fee, LTV, Net Advance)



[Gemini Structured Credit Memo]



[Consolidated JSON Response]

—(200 OK)—> [Dynamic React State]